

Control and Coordination

Science • Chapter 6 • Chapter Summary & Study Guide for Daksh

When a cyclist swerves to avoid a pothole, when our pupil shrinks in bright sunlight, when a plant tilts towards the window — all these are examples of **control and coordination**. Living things constantly sense their environment and respond. Animals use a fast, electrical **nervous system** and a slower, chemical **endocrine (hormonal) system**. Plants, lacking nerves, rely entirely on chemical signals — **plant hormones**.

The nervous system in animals

The unit cell is the **neuron**: a cell body with dendrites (receive signals) and a long axon (carries the signal away). Signals are electrical along the neuron and **chemical** at the gap between two neurons — the **synapse** — where neurotransmitters are released.

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Stimulus -> Receptor -> Sensory neuron -> SPINAL CORD / BRAIN ->
                                         Motor neuron -> Effector (muscle/gland) -> Response
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REFLEX ARC (no waiting for the brain): the spinal cord shortcut
e.g. hand jerks back from a hot plate before you 'think'.
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The human nervous system is divided into:

- **Central Nervous System (CNS)** — brain + spinal cord.
- **Peripheral Nervous System (PNS)** — all the nerves connecting CNS with the body.
- Brain has 3 main parts: **fore-brain** (cerebrum — thinking, voluntary action, memory; hypothalamus — homeostasis, hunger, sleep), **mid-brain** (relays signals), **hind-brain** (cerebellum — balance and posture; medulla — involuntary actions like breathing, heart-beat; pons — sleep regulation, breathing rhythm).
- **Reflex actions** are quick, involuntary responses controlled mainly by the spinal cord — they bypass the brain and save crucial milliseconds.
- **Voluntary actions** (writing, walking) are controlled by the cerebrum.

How a muscle moves on receiving a signal

When the signal reaches a muscle cell, special proteins change shape. The cell shrinks (contracts), pulling on the bone it's attached to and producing the movement. This is why we can move our arms within a fraction of a second of deciding to.

Hormones (Endocrine system) in animals

Hormones are **chemical messengers** released by ductless (endocrine) glands directly into the bloodstream. They reach target organs and produce specific effects. Important glands and their hormones:

- **Pituitary** (master gland, in brain) — Growth hormone (controls body growth — too little gives dwarfism, too much gigantism).
- **Thyroid** — Thyroxine (regulates metabolism; needs iodine — deficiency causes goitre).
- **Pancreas** — Insulin (regulates blood sugar; deficiency causes diabetes).
- **Adrenals** — Adrenaline (released during stress: raises heart rate, breathing — the 'fight or flight' hormone).
- **Testes** — Testosterone; **Ovaries** — Oestrogen, Progesterone (sex hormones — drive puberty and reproduction).

Feedback regulation: the level of a hormone is automatically controlled — e.g. when blood sugar rises after a meal, pancreas releases more insulin; when sugar falls, insulin secretion is reduced.

Coordination in plants

Plants don't have a nervous system. They respond by: **movement** (immediate, e.g. touch-me-not folding) — caused by changes in turgor pressure of cells; **growth-related responses** (slow) — caused by plant hormones.

- **Tropisms** — directional growth in response to a stimulus.

Phototropism (light), Geotropism/gravitropism (gravity), Hydrotropism (water), Chemotropism (chemicals — e.g. growth of pollen tube towards ovule).

- **Plant hormones:** **Auxins** (cell elongation; bend toward light), **Gibberellins** (stem growth, fruit/seed development), **Cytokinins** (cell division), **Abscisic acid** (inhibits growth, closes stomata in stress, induces dormancy).

★ MUST-REMEMBER POINTS

- Neuron's three parts: dendrite (receive), cell body, axon (transmit).
- Synapse = chemical junction (neurotransmitters); signals go in only one direction.
- Reflex arc = receptor → sensory neuron → spinal cord → motor neuron → effector.
- Cerebrum = thinking; Cerebellum = balance; Medulla = involuntary (heartbeat, breathing).
- Pituitary is master gland; Insulin from pancreas (low → diabetes); Adrenaline = stress hormone.
- Plants — auxin (elongation, phototropism), gibberellin (stem growth), cytokinin (cell division), ABA (inhibits growth).
- Tropisms: phototropism (light), geotropism (gravity), hydrotropism (water), chemotropism (chemicals).

— End of Summary —